JNTUGV-CEV — Department of Information Technology

DATA SCIENCE WITH PYTHON — B.Tech Course Syllabus (Revised)

Course Objectives:

- Provide B.Tech students with the knowledge and expertise to become a proficient data scientist.
- Demonstrate an understanding of statistics and machine learning concepts that are vital for data science.
- Produce Python code to statistically analyze a dataset.
- Critically evaluate data visualizations based on their design and use for communicating stories from data.
- Build, train and evaluate basic machine learning models for real-world data science problems.

Course Outcomes:

		Knowledge Level (K)#
CO1	Explain how data is collected, managed and stored for data science.	K2
CO2	Understand the key concepts in data science, including their real-world applications and the toolkit used by data scientists.	K1
CO3	Discuss different Python packages (NumPy, Pandas, Matplotlib) and how to use them for data analysis.	K5
CO4	Implement data collection, preprocessing and visualization scripts using Python Pandas and Matplotlib.	K6
CO5	Build and evaluate basic machine learning models for data science applications using Python.	K6

#Based on suggested Revised BTL

Mapping of Course Outcomes with Program Outcomes

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	2	2	2	2	3	2
CO2	1	2	3	3	3	2
CO3	1	1	2	3	3	1
CO4	2	1	2	3	3	1
CO5	2	2	3	3	2	1

(Please fill the above with Levels of Correlation, viz., L, M, H)

UNIT	**CONTENTS**	**Contact Hours**
UNIT – 1	**PYTHON Basics and Programming Concepts:** Introducing Python, Types and Operations – Numbers, Strings, Lists, Tuples, Dictionaries, Files, Numeric Types, Dynamic Typing; Statements and Syntax – Assignments, Expressions, Statements,	10 Hrs

	Loops, Iterations, Comprehensions; Functions – Function Basics, Scopes, Arguments, Advanced Functions; Modules – Module Coding Basics, Module Packages, Advanced Module Topics; Classes and OOP – Class, Operator Overloading, Class Designing; Exceptions and Tools – Exception Basics, Exception Coding Details, Exception Objects, Designing With Exceptions, Parallel System Tools.	
UNIT – 2	**Python Tools for Data Handling: GUI, APIs, Web Data and Databases:** Python GUI Programming Basics – Introduction to GUI development options, building simple interfaces with Tkinter widgets; Working in Jupyter Notebook/IPython as the data science environment; Internet Programming – Network Scripting basics, working with REST APIs using the requests library (JSON responses, authentication basics); Web Scraping basics using BeautifulSoup; Databases and Persistence – SQL fundamentals, connecting Python to databases using sqlite3/SQLAlchemy; working with data structures, text and language for data handling.	10 Hrs
UNIT – 3	**Pandas and NumPy:** Numpy Basics – Fast Element wise array functions, Multidimensional Array, Data Processing using arrays, file i/o with arrays; Pandas – Data Structures, Essential Functionality, Summarizing and Computing Descriptive Statistics, Handling Missing Data, Hierarchical Indexing.	12 Hrs
UNIT – 4	**Data Preprocessing and Visualization:** Data Loading, Storage, and File Formats – Reading and Writing data in text format, binary data formats, interacting with html and web APIs, interacting with databases; Data Wrangling: Clean, Transform, Merge, Reshape – Combining and Merging Data Sets, Reshaping and Pivoting, Data Transformation, String Manipulation; Data Aggregation and Group Operations – Group by Mechanics, Data Aggregation, Groupby Operations and Transformations, Pivot Tables and Cross Tabulation; Data Visualization – A Brief matplotlib API Primer, Statistical Visualization with Seaborn, Plotting Functions in pandas, Time Series, real-world case study applications.	12 Hrs
UNIT – 5	**Machine Learning for Data Science:** Introduction to Machine Learning – Types of Learning: Supervised, Unsupervised and Reinforcement Learning, the Machine Learning workflow; Data Preparation for ML – Feature Engineering, Feature Scaling, Train-Test Split, Cross-Validation; Supervised Learning – Linear Regression, Logistic Regression, k-Nearest Neighbours, Decision Trees; Unsupervised Learning – k-Means Clustering, Hierarchical Clustering, Dimensionality Reduction with PCA; Model Evaluation – Overfitting and Underfitting, Confusion Matrix, Accuracy, Precision, Recall, F1-Score, ROC Curve; Introduction to scikit-learn – building, training and evaluating ML models in Python.	12 Hrs
	Total	**56 Hrs**

Text Books:

1. Learning Python, O'Reilly, Mark Lutz
2. Programming Python, O'Reilly, Mark Lutz
3. Python for Data Analysis, O'Reilly, Wes McKinney
4. Introduction to Machine Learning with Python, O'Reilly, Andreas C. Müller & Sarah Guido

Reference Books:

1. Python: The Complete Reference, Martin C. Brown, McGraw Hill Education

2. Head First Python, Paul Barry, O'Reilly
3. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow, Aurélien Géron, O'Reilly